

Day 3 - Lab Exercises

Prompt-to-Program Lab

Sequence – sequence is line 1 runs, then line 2 and then line 3 in sequence.

Selection – makes decisions inside a loop like if/else

Iteration – loops through items such as a for loop, while loop or forEach loop.

You can use them separately or all together. Such as going through a list of test scores, for each score above 70 is a passing score, otherwise say needs more practice. Which uses all three.

Exercise 1. Build with Sequence: Screen recording added

Verify code after prompt: Yes, the sequence makes sense. The line depends on the previous line. Hourly pay must come first. Next, hours worked and weekly pay cannot be calculated unless Hourly pay and hours worked must come first.

Exercise 2. Build with Selection: Screen recording added

If I add a negative number, it defaults to the last option if it doesn't meet any of these conditions then select the last option.

Big Hunt Exercise 2: The logic Error

Yes the bug exists in this example. The variable is wrong. The variable is 85 making every score 85 and 85 can't be larger than 90.

Create an error, Make score 85 an A:

```
<!DOCTYPE html>
```

```
<html>
```

```
<body>
```

```
<script>
```

```
// Ask the user for their score
```

```
const score = parseFloat(prompt("Enter your score:"));

// Determine the grade and display the result
if (score >= 82) {
    document.write("<p>You passed! Grade: A</p>");
} else if (score >= 80 && score <= 89) {
    document.write("<p>You passed! Grade: B</p>");
} else if (score >= 70 && score <= 79) {
    document.write("<p>You passed! Grade: C</p>");
} else {
    document.write("<p>You failed with a score of " + score + "</p>");
}

</script>
</body>
</html>
```

Bug Hunt Exercise 2: The Missing Piece:

Problem Solved:

```
const items = [
    { name: "Apples", price: 3.99 },
    { name: "Bread", price: 2.49 },
    { name: "Milk", price: 4.29 },
    { name: "Eggs", price: 5.99 }
];
```

```
let total = 0;
```

```
for (let i = 0; i < items.length; i++) {  
    document.write("<p>" + items[i].name + ": $" + items[i].price + "</p>");  
  
    total = total + items[i].price;  
}
```

```
let average = total/items.length;  
document.write("<h3>Total: $" + total.toFixed(2) + "</h3>");  
document.write("<h3>Average: $" + average.toFixed(2) + "</h3>");
```

Extend Challenge: Combine Everything -- Level 3

Build a program that uses sequence, selection, AND iteration:

Write a JavaScript program that has a list of 5 students with their test scores. Loop through the list. For each student:

- If their score is 90 or above, display "[Name]: A - Excellent!"
- If 80-89, display "[Name]: B - Good job!"
- If 70-79, display "[Name]: C - Keep working!"
- Below 70, display "[Name]: Needs improvement"

At the end, display how many students got an A.

```
const students = [  
    {name: "Joan", score: 99},  
    {name: "Michael", score: 80},  
    {name: "Angie", score: 75},  
    {name: "Shawn", score: 95},  
    {name: "Darla", score: 89 }  
];  
  
for (let i = 0; i < students.length; i++) {  
    const student = students[i];
```

```
let grade = "";

if (student.score >= 90) {
    grade = "Excellent";
} else if (student.score >= 80) {
    grade = "Great";
} else if (student.score >= 70) {
    grade = "Good Job";
} else if (student.score >= 60) {
    grade = "Fair";
} else {
    grade = "F";
}

if (grade === "F") {
    document.write("<p>" + student.name + ": You failed with a " + student.score +
"</p>");
} else {
    document.write("<p>" + student.name + ": You passed! (" + grade + ") Score: "
+ student.score + "</p>");
}
}
```